inhabitant, produces Tetanus when it is introduced into a wound or an abrasion. Tetanus is thus an infectious disease, but not communicable.

Infectious disease occurs when the infecting microorganism causes damage to the host. The term infection refers to the establishment of the microorganisms in the tissues resulting in injury or harmful effect to the host. Infection is a pathological condition due to the growth of microorganisms in a host. To initiate an infection, a pathogenic microbe enters the tissues of the body by a characterization route, the portal of entry.

12.1.1 Routes of Infections

There are various ways in which microorganisms enters into the host are explained below.

a. Contact

Infection may be acquired by contact which may be direct or indirect. Sexually transmitted diseases such as syphilis and gonorrhea spread by direct contact. Indirect contact may be through the agency of inanimate objects such as clothing, pencils or toys which may be contaminated by a pathogen from one person to another. Pencils shared by school children may act as fomites in the transmission of diphtheria, and face towels in trachoma.

b. Inhalation

Respiratory infections such as influenza and tuberculosis are transmitted by inhalation of the pathogen in droplet and droplet nuclei that are shed by the patients during sneezing, speaking or coughing. Common cold virus, Adenovirus is some of the virus producing respiratory infections.

c. Ingestion

Intestinal infections are generally acquired by the ingestion of food or drinks contaminated by pathogens. Infection transmitted by ingestion may be waterborne (cholera), food borne (typhoid) or fecal-oral route (dysentery).

d. Inoculation

Pathogens, in some instances, may be inoculated directly into the tissues of the host. Tetanus spores implanted in the depth of wounds, rabies virus deposited subcutaneously by dog bites, inoculation through unsterile syringes and surgical equipments are examples that enter through direct inoculation.

e. Congenital

Some pathogens are able to cross the placental barrier and infect the fetus in uterus. Bacteria like *Treponema pallidum*, viruses like *Rubella, Cytomegalovirus* parasite like *Toxoplasma gondi* are some of the organisms that enter through placenta and cause disease in the newborn.

12.1.2 Types of Infections

Infections may be classified in various ways. Initial infection with a parasite in a host is called a primary infection. Subsequent infections by the same parasite in the host are termed reinfections. When a new parasite sets up an infection in a host whose resistance is lowered by a preexisting infectious disease, this is termed secondary infection.

When in a patient already suffering from a disease, a new infection is setup

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